

The root number of the tensor tower of an elliptic curve: a closed form, a Kummer criterion, and odd-order central vanishing at the Fermat grades

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Abstract

Let $E/\mathbb{Q}$ be an elliptic curve and $M_g = H^1(E)^{\otimes g}$ the g -fold tensor motive. We give the archimedean epsilon factor of the tower in closed form: with $A(g) = 2^{g-1} + g \binom{g-1}{\lfloor (g-1)/2 \rfloor}$ one has $\varepsilon_\infty(M_g) = i^{A(g)}$, the exponent $A(g)$ is even at every grade (the sign is real), $\varepsilon_\infty = +1$ at every even grade, and for odd $g \geq 3$

$$\varepsilon_\infty(M_g) = -1 \iff g = 2^k + 1,$$

by Kummer’s carry criterion for the 2-adic valuation of a central binomial coefficient. For semistable E we then assemble the *global* root number of $L(M_g, s)$ across the Clebsch–Gordan decomposition and prove that at every Fermat grade $g = 2^k + 1$ ($k \geq 2$) the global sign is -1 *for both parities of $\varepsilon(E)$* : $L(M_g, s)$ vanishes at $s = (g+1)/2$ to *odd* order at least $M(g, \varepsilon(E))$, an explicitly computed ledger (9 and 1 at $g = 5$ for the two parities; 91 and 35 at $g = 9$; $\approx 1.9 \times 10^{18}$ by $g = 65$). At $g = 3$ the ledger recovers the Gross–Schoen grade ($M = 3$ for $\varepsilon(E) = -1$), matching the sign of the triple-product L -function (Gross–Kudla); above $g = 3$ the tensor tower’s sign appears not to have been computed before. The combinatorial laws are machine-checked (exact ledger values through $g = 9$ and sign parities through $g = 25$ in the kernel); the arithmetic inputs are the classical local–global sign computations of Deligne, Martin–Watkins, Dummigan–Martin–Watkins and Saito, and the functional equations are unconditional by symmetric-power functoriality (Newton–Thorne).

1 The object and what is known

Write $V = H^1(E)$, a motive of weight 1 with $\det V = \mathbb{Q}(-1)$, and $M_g = V^{\otimes g}$, of weight g and rank 2^g . Its L -function factors through the Clebsch–Gordan decomposition

$$V^{\otimes g} = \bigoplus_{j=0}^{\lfloor g/2 \rfloor} (\mathrm{Sym}^{g-2j} V)(-j)^{\oplus m_{g,j}}, \quad m_{g,j} = \binom{g}{j} - \binom{g}{j-1}, \quad (1)$$

the multiplicities being the ballot numbers (the decomposition holds at Chow-motive level [1]), so that $L(M_g, s) = \prod_j L(\mathrm{Sym}^{g-2j} E, s - j)^{m_{g,j}}$. Every twisted factor’s completed functional

*With machine-checked components in Lean 4 (`RootNumberLaw.lean`, `TensorTowerGlobalSign.lean`); axiom footprint `{propext, Classical.choice, Quot.sound}`, no `sorry`.

equation reflects about the same axis: the center of the piece $\mathrm{Sym}^{g-2j}(-j)$ sits at $s = \frac{g-2j+1}{2} + j = \frac{g+1}{2}$, the center of the tensor motive. By symmetric-power functoriality for holomorphic newforms [7] (non-CM; the CM case is classical via Hecke characters) each factor is automorphic, so $L(M_g, s)$ has entire continuation (odd g) and its functional equation is a theorem, not a hypothesis.

The literature on the *sign* of the tower stops at $g = 3$. At $g = 1$ the sign theory is classical. At even g the sign is $+1$: for $g = 2$ by the symplectic $\times$ symplectic mechanism (Prasad–Ramakrishnan [9, Prop. 2.1], the argument due to Rohrlich [10]; cf. [9, Prop. 6.1] for $\mathrm{Sym}^2 \otimes \mathrm{Sym}^2$), and for even $g \geq 4$ via (1) from the local $+1$ at every even symmetric power [4, Prop. 7.1] with Saito’s global theorem [11]. At $g = 3$ the global sign of the triple product is computed by Gross–Kudla [5] for newforms of common square-free level (their (1.9) is exactly the Hodge multiset of $V^{\otimes 3}$, and the leading -1 of their sign formula is the archimedean factor), with the local archimedean dichotomy in Prasad [8]. For $g \geq 4$ we are not aware of any computation of the sign of the tensor tower, nor of any appearance of 2-adic binomial criteria in the root-number literature.

2 The archimedean law

Deligne’s theory of local constants [2], in the form of the archimedean recipe [3, §5.3], assigns to a Hodge structure its archimedean epsilon factor through the Hodge types. Applied to (1) — or directly to the Hodge diamond of M_g , whose type- (p, q) multiplicity is $\binom{g}{p}$ with $p + q = g$ — the exponent of i is

$$A(g) = \sum_{p < g/2} (q - p + 1) \binom{g}{p} + \frac{1}{2} \binom{g}{g/2} \Big|_{g \text{ even}}.$$

For odd g no (p, p) block occurs, so everything below is independent of the convention for the balanced middle block.

Theorem 1 (archimedean sign law; machine-checked). *For all $g \geq 1$: (i) $A(g) = 2^{g-1} + g \binom{g-1}{\lfloor (g-1)/2 \rfloor}$; (ii) $A(g)$ is even, so $\varepsilon_\infty(M_g) = i^{A(g)} \in \{\pm 1\}$; (iii) $\varepsilon_\infty = +1$ for every even g ; and for odd $g \geq 3$,*

$$\varepsilon_\infty(M_g) = -1 \iff A(g) \equiv 2 \pmod{4} \iff g = 2^k + 1, \quad k \geq 1.$$

Proof (sketch; full proof in Lean). (i) The deviation part $\sum_p (g-2p) \binom{g}{p}$ telescopes to $g \binom{g-1}{\lfloor (g-1)/2 \rfloor}$ and the flat part is a half row, 2^{g-1} . (ii) is the parity of a central binomial. (iii) For odd $g = 2m+1$ the condition $A \equiv 2 \pmod{4}$ reduces to $v_2 \binom{2m}{m} = 1$, and by Legendre’s formula $v_2 \binom{2m}{m} = s_2(m)$, the binary digit sum; $s_2(m) = 1$ iff m is a power of two, i.e. $g = 2^k + 1$. This is Kummer’s carry criterion: the sign flips exactly when adding $m+m$ in base 2 carries once. The Lean formalisation (`RootNumberLaw.lean`) proves (i)–(iii) unconditionally as statements about A ; the identification of A with Deligne’s exponent is the cited classical recipe. As an independent check, assembling the per-piece archimedean signs $\varepsilon_\infty(\mathrm{Sym}^m) = -\binom{-2}{m}$ [3, §5.3] across (1) reproduces $i^{A(g)}$ at $g = 3, 5, 7$ and the closed form through $g = 11$. $\square$

The even half of Theorem 1 recovers what the Stiefel–Whitney mechanism knows [9, 10]; the odd half is, to our knowledge, new above $g = 3$, where it recovers Gross–Kudla’s -1 .

3 The global law for semistable curves

For semistable $E/\mathbb{Q}$ and odd m the global sign of $L(\text{Sym}^m E, s)$ is

$$w_m = \left(\frac{-2}{m}\right) \varepsilon(E), \quad (2)$$

assembled from Deligne's archimedean factor and $w_m(p) = w_1(p)^m$ at multiplicative p [6, §4.3], [4]; even powers have sign $+1$ [11]. The sign of a factor is unchanged by the shift to the common center (the Hodge gaps $q - p$ are unchanged, and the unramified twist scales the finite epsilon factors by a positive real), so the global sign of the tensor tower is the ballot-weighted product of (2), and each factor with $w = -1$ vanishes to odd order at the common center. Define the *forced multiplicity ledger*

$$M(g, \varepsilon_E) = \sum_{j: w_{g-2j} = -1} m_{g,j}, \quad \text{so} \quad \text{ord}_{s=\frac{g+1}{2}} L(M_g, s) \geq M(g, \varepsilon_E),$$

and the global sign is $(-1)^{M(g, \varepsilon_E)}$.

Theorem 2 (Fermat grades force vanishing for every semistable curve). *Let $E/\mathbb{Q}$ be semistable and $g = 2^k + 1$ with $k \geq 2$. Then $M(g, +1)$ and $M(g, -1)$ are both odd: the global root number of $L(M_g, s)$ is -1 regardless of $\varepsilon(E)$, and $L(M_g, s)$ vanishes at $s = (g + 1)/2$ to order at least $M(g, \varepsilon(E)) \geq 1$.*

Proof. Write $S_- = \sum_{\left(\frac{-2}{g-2j}\right)=-1} m_{g,j}$ and S_+ for the complementary sum, so $M(g, +1) = S_-$, $M(g, -1) = S_+$, and by the telescoping of the ballot numbers $S_+ + S_- = \binom{g}{(g-1)/2}$. For $k \geq 3$ we have $g \equiv 1 \pmod{8}$, and as j increases the class of $g - 2j \pmod{8}$ cycles $(1, 7, 5, 3)$, so the $\left(\frac{-2}{\cdot}\right) = -1$ residues occupy the consecutive pairs $j \equiv 1, 2 \pmod{4}$. Within each pair the ballot numbers telescope: $m_{g,4t+1} + m_{g,4t+2} = \binom{g}{4t+2} - \binom{g}{4t}$, and since $(g - 1)/2 = 2^{k-1} \equiv 0 \pmod{4}$ the pairs are complete. Hence

$$S_- \equiv \sum_{\substack{0 \leq j \leq 2^{k-1}-2 \\ j \text{ even}}} \binom{g}{j} \pmod{2}.$$

By Lucas' theorem at $p = 2$, $\binom{2^k+1}{j}$ is odd iff the binary digits of j are dominated by $10 \cdots 01_2$, i.e. $j \in \{0, 1, 2^k, 2^k + 1\}$; in the range only $j = 0$ contributes, so $S_- \equiv 1$. The total $\binom{g}{2^{k-1}}$ is even by Kummer (adding 2^{k-1} and $2^{k-1} + 1$ carries at bit $k - 1$), so $S_+ = \binom{g}{2^{k-1}} - S_-$ is odd as well. The case $k = 2$ ($g = 5$) is a direct kernel check. Each sign- (-1) factor $L(\text{Sym}^{g-2j} E, s - j)$ vanishes to odd order ≥ 1 at the common center and occurs $m_{g,j}$ times. $\square$

The sign behaviour is sharp. Central vanishing itself is generic here: $M(g, \varepsilon_E) > 0$ for every odd $g \geq 5$ and both parities. What distinguishes the Fermat grades is *odd-order* vanishing — global sign -1 — for both parities at once (sign parities kernel-checked through $g \leq 25$; the grades $g = 33, 65$ are covered by Theorem 2):

g	$M(g, -1)$	$M(g, +1)$	global sign -1 for
3	3	0	$\varepsilon(E) = -1$ only (Gross–Schoen grade)
5	9	1	both (Fermat) — incl. curves with no sign-forced vanishing below Sym^5
7	28	7	$\varepsilon(E) = +1$ only
9	91	35	both (Fermat)
13	1078	638	neither
17	14179	10131	both (Fermat)
25	2848366	2351934	neither
33	623067171	543735939	both (Fermat)
65	1863283301142158627	1746430915865974243	both (Fermat)

Corollary 3. *For every semistable $E/\mathbb{Q}$ and every $k \geq 2$, the tensor-tower L -function $L(H^1(E)^{\otimes(2^k+1)}, s)$ vanishes at its center to odd order: the global root number is -1 regardless of $\varepsilon(E)$. Since central vanishing of even order occurs generically at odd grades, the Fermat grades are exactly the displayed grades where the sign forces vanishing for every semistable curve at once. Under the Beilinson–Bloch philosophy each forced odd-order zero predicts null-homologous cycles on E^{2^k+1} generalising the Gross–Schoen modified diagonal (the $g = 3$ row, where the forcing occurs for $\varepsilon(E) = -1$ and $M = 3$ matches the classical triple-product case [5]).*

Remark 4 (what is new, what is cited). *The closed form of Theorem 1, the Kummer characterisation, the ballot assembly of Theorem 2 and its Fermat-parity law appear to be new; the per-piece signs (2), Deligne’s recipe, Saito’s even-power theorem and the Newton–Thorne functional equations are the cited inputs. The combinatorial statements are machine-checked in Lean 4 at standard axiom footprint: the closed form, the parity and digit laws, the ballot telescoping, the exact ledger values at $g = 3, 5, 9$, and the sign parities at $g = 7, 13, 17, 25$ are kernel (**decide**) theorems. The parities at $g = 33, 65$ require no computation — they are Fermat grades, covered by Theorem 2 — and the displayed magnitudes at $g = 13, 25, 17, 33, 65$ are evaluator outputs of the same compiled definition (the $g = 13$ values independently hand-verified in audit). The forced multiplicities are lower bounds from sign parity alone; no part of this note assumes any conjecture.*

References

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